

Special products



- 1 Determine whether the following statements are true. If not, rewrite the right-hand side to make the statement true.

a $(8 + 6)^2 = 8^2 + 6^2$

b $(y - 7)^2 = y^2 - 21y + 49$

- 2 Complete each of the following statements:

a $(x + 3)^2 = x^2 - 1 + 9$

b $(x - 3)^2 = x^2 - 1 + 1$

c $x^2 + 64 - 1 = (x - 8)^2$

d $(x + 1)^2 = x^2 + 1 + 36$

e $(8x^1 - 9)^2 = 64x^4 - 144x^2 + 81$

f $(-9x + 2)^2 = 81x^2 - 1 + 1$

g $(2y + 1)(2y - 1) = 4y^2 - 4$

h $(3x - 8)^2 = 9x^2 - 1 + 64$

3

- a Find the product $(t + 5)^2$.

- b Find the product $(-t - 5)^2$.

- c Determine whether the following are equal to $(-t - 5)^2$.

i $t^2 - 10t + 25$

ii $(-t)^2 + 10t + (-5)^2$

iii $-(t + 5)^2$

iv $(t + 5)^2$

- 4 Find the product of each of the following:

a $(x + 3)^2$

b $(m - 5)^2$

c $(s + 4t)^2$

d $(x^2 + 3)^2$

e $(9x + 2)^2$

f $(4 - 5m)^2$

g $(4x + 7y)^2$

h $3(3x + 8y)^2$

i $9(3x + 8y)^2$

j $3x(7x - 5y)^2$

k $\left(\frac{8x - 9}{9}\right)^2$

l $\left(6y + \frac{1}{2}\right)^2$

m $\left(\frac{n}{4} - \frac{4}{n}\right)^2$

5 Find the product of each of the following:



a $(m + 3)(m - 3)$

c $(3x - 8)(3x + 8)$

e $(4y + 9)(4y - 9)$

g $(-7y - 8)(-7y + 8)$

i $(10 - m)(m + 10)$

k $5x(2x - 9y)(2x + 9y)$

m $(9x - 0.4)(9x + 0.4)$

o $\left(v + \frac{1}{7}\right)\left(v - \frac{1}{7}\right)$

q $\left(6x - \frac{8}{3}\right)\left(6x + \frac{8}{3}\right)$

s $a^2 + (11 - a)(11 + a) - 10$

b $(x - 7)(x + 7)$

d $(4x - 3y)(4x + 3y)$

f $(6 - p)(6 + p)$

h $(11 - 9p)(11 + 9p)$

j $6(8x - 9y)(8x + 9y)$

l $(0.8x + 9)(0.8x - 9)$

n $(0.2x - 11)(0.2x + 11)$

p $\left(6x + \frac{1}{3}\right)\left(6x - \frac{1}{3}\right)$

r $\left(x - \frac{2}{x}\right)\left(x + \frac{2}{x}\right)$

6 Evaluate 403^2 by expressing 403 as the sum of 400 and 3, and then evaluating the product. Show your work.

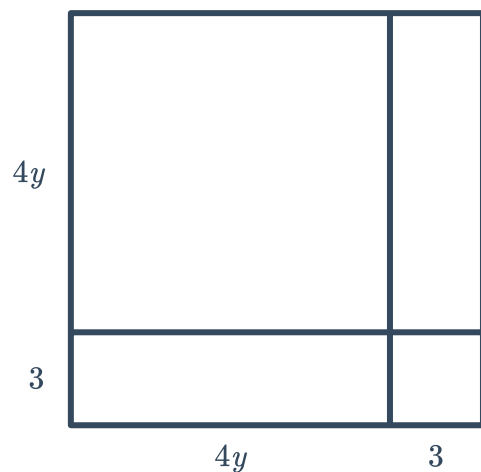
7 Evaluate 896^2 by expressing 896 as $900 - 4$, and then evaluating the product. Show your work.

8 Find the value of $201 \cdot 199$ by rewriting it in the form $(x + y)(x - y)$. Show your work.

9 Consider the following figure.

a Find the area of the large square by multiplying its side lengths.

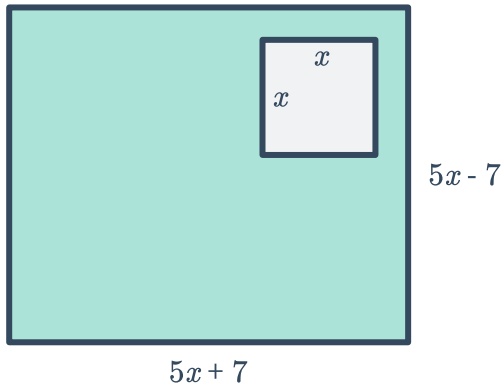
b Find the area of the large square by adding the areas of the smaller rectangles inside it.



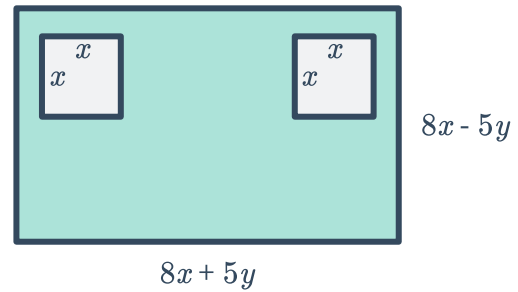
10 For each of the following diagram, form an expression for the shaded area.



a

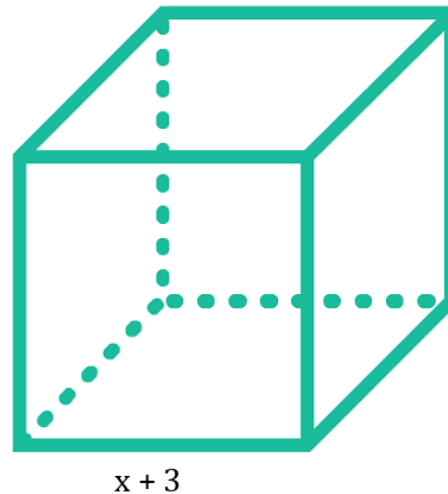


b



11 Consider the following cube.

- a Find the surface area of the cube.
- b Find the volume of the cube.



12 Write down an expression for the product of three consecutive integers if the middle integer is m .

13

- a Find the product $(y^2 + 9)(y^2 - 9)$.
- b Find an expression for the total distance traveled when traveling at $\frac{1}{y}(y^2 + 9)$ kilometers per hour for $y^2(y^2 - 9)$ hours.



Additional questions

14 Prove the following results about special products:

a $(a + b)^2 = a^2 + 2ab + b^2$

b $(a - b)^2 = a^2 - 2ab + b^2$

c $(a + b)(a - b) = a^2 - b^2$

15 Show how you can use one of the special products $(a \pm b)^2$ to find each perfect square without a calculator.

a 21^2

b 19^2

c 33^2

d 47^2

16 A square with side lengths measuring $x - 1$ centimeters has each side enlarged by a factor of 2. Find an expression, for the area of the new square.

17 Show how the difference of squares pattern can be used to make the problem $105^2 - 104^2$ easier to evaluate without the use of a calculator.

18 Polynomial multiplication can be used to find the total area of a rectangle that has a border, where the length and width each represent a factor in the product.

a Write an algebraic model that can be used to find the total area of a square with an unknown side length and a border with an unknown border width.

b Use your model to find the total area of a square with a side length of $2m + 1$ and a border width of 3